

REGULARITY THEORY IN FREE BOUNDARY PROBLEM

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1. INTRODUCTION

1.1. Basic settings and existence. In this course, we focus on the Alt-Phillips problem. Given a bounded domain $\Omega \subset \mathbb{R}^d$ and $\gamma \in (-2, 2)$, the Alt-Phillips energy is

$$E_\gamma(u, \Omega) := \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 + u^\gamma \chi_{\{u>0\}} \right) dx. \quad (1.1)$$

Note that these two terms $\frac{1}{2} |\nabla u|^2$ and $u^\gamma \chi_{\{u>0\}}$ have a competing relation. Note that for a given non-trivial boundary condition, to make the energy from the second term small, u should be 0. However, the non-vanishing boundary condition will cause the largeness of the energy term $\int_{\Omega} |\nabla u|^2$ since if u decays too fast to 0, the derivative of u contributes more to the whole energy functional. In all, to minimize (1.1), one needs to combine these two phenomena. Before we delve into specific topics, we first give some general classification of different problem settings. For simplicity, we call $\frac{1}{2} \int_{\Omega} |\nabla u|^2$ as the diffusion term and $\int_{\Omega} u^\gamma \chi_{\{u>0\}}$ as the reaction term, respectively.

- For $\gamma \geq 2$, the diffusion term is in the dominant place, and the solution of the minimizing problem for (1.1) behaves like a solution of a semilinear elliptic equation.
- If $\gamma = 1$, it is the well-known obstacle problem.
- If $\gamma = 0$, we call it the Bernoulli problem in Fluid dynamical theory and the Alt-Caffarelli problem in functional analysis.
- If $\gamma = -2$, minimizing solutions of (1.1) correspond to the minimal surfaces.

Remark 1.1. Note that for $\gamma \geq 1$, the reaction term in the Alt-Phillips energy is convex, and the energy functional is convex.

In this course, our primary focus is on the case that $\gamma \in [0, 2)$. Without extra explanations, we always have such an assumption on the range of γ . To begin with, we give the existence of minimizers of Alt-Phillips energy under basic settings, as we always want our set of research elements to be non-empty.

Theorem 1.2 (Existence). *Let $\gamma \in [0, 2)$. Given a Lipschitz domain $\Omega \subset \mathbb{R}^d$, and a Lipschitz function (possibly enough if only bounded) $g : \partial\Omega \rightarrow [0, +\infty)$, we have*

$$u \in H_g^1(\Omega) := \{v \in H^1(\Omega) : v = g \text{ on } \partial\Omega\}$$

such that for any $v \in H_g^1(\Omega)$,

$$E_\gamma(u, \Omega) \leq E_\gamma(v, \Omega).$$

In particular, the Alt-Phillips energy attains its infimum at the function u .

Proof. We apply the direct method. Let

$$E := \inf_{v \in H_g^1(\Omega)} E_\gamma(v, \Omega).$$

There is a minimizing sequence $\{u_n\}_{n \in \mathbb{Z}_+} \subset H_g^1(\Omega)$ such that $E_\gamma(u_n, \Omega) \rightarrow E$ as $n \rightarrow +\infty$. Without loss of generality, we assume that $0 \leq u_n \leq \sup_{\partial\Omega} g$ in Ω . If not, we consider a new sequence

$$\bar{u}_n := \min \left\{ \max\{u_n, 0\}, \sup_{\partial\Omega} g \right\}.$$

Note that for sufficiently large n ,

$$\|\nabla u_n\|_{L^2(\Omega)}^2 \leq E_\gamma(u_n, \Omega) \leq E + 1,$$

it follows that $\|u_n\|_{H^1(\Omega)}$ is uniformly bounded. Then, up to a subsequence, $u_n \rightharpoonup u$ weakly $H^1(\Omega)$ and $u_n \rightarrow u$ a.e. in Ω . By the dominant convergence theorem, we have

$$\int_{\Omega} u^\gamma \chi_{\{u>0\}} dx = \lim_{n \rightarrow +\infty} \int_{\Omega} u_n^\gamma \chi_{\{u_n>0\}} dx. \quad (1.2)$$

Since $\|\cdot\|_{L^2}$ is lower semi-continuous with respect to the weak convergence, we obtain that

$$\int_{\Omega} \|\nabla u\|^2 \leq \liminf_{n \rightarrow +\infty} \int_{\Omega} \|\nabla u_n\|^2.$$

This, together with (1.2), implies that

$$E_\gamma(u, \Omega) \leq \liminf_{n \rightarrow +\infty} E_\gamma(u_n, \Omega). \quad (1.3)$$

Note that the weak convergence of u_n in $H^1(\Omega)$ ensures the strong convergence in $L^2(\partial\Omega)$. As a result, $u = g$ on $\partial\Omega$. Then, $u \in H_g^1(\Omega)$. Combining (1.3), we have $E_\gamma(u, \Omega) = E$ and completes the proof. $\square$

1.2. Main problem and review of strategies. To extend our issues in this course, we present the main question and methods to be applied.

Definition 1.3. Let $\{u > 0\}$ be the positive set and define $\{u = 0\}$ as the contact set. We call $\partial\{u > 0\}$ as the free boundary.

Question 1.4 (Main question). Let $\gamma \in [0, 2)$. Assume that $u \in H^1(\Omega)$ is a minimizer (1.1)¹ of the Alt-Phillips energy. The problem is about the regularity of the free boundary $\partial\{u > 0\}$.

We will first sketch the strategies and frameworks in the study of the regularity of the free boundary.

Stage 1 Let $0 \in \partial\{u > 0\}$, we consider the blow-up sequence $u_r(x) = r^{-\beta} u(rx)$, where $\beta > 0$ is to be determined later. We mainly consider three issues.

- The compactness of the blow-up sequence. We need to establish the convergence properties of the blow-up limits.
- The no degeneracy of the blow-up limits. We intend to ensure that the blow-up limits are non-vanishing.
- The simplified properties of the blow-up limits. Indeed, in the later analysis, we will prove that the blow-up limits are all minimizers and homogeneous under some scaling. Such a property implies that the limit of the blow-up is simpler than the original minimizer since it has only $(d-1)$ free variables.

Stage 2 We consider the classification of the set of minimal cone:

$$K(\gamma, d) := \{\text{homogeneous minimizers of } E_\gamma(\cdot, \mathbb{R}^d)\}.$$

By analyzing the set $K(\gamma, d)$, one can classify the blow-up limits and obtain more information on the minimizing problem.

¹Here we call u is a minimizer if u is a minimizer under some specific boundary condition as in Theorem 1.2. Since we always consider the interior problem, we do not emphasize the boundary condition.

Stage 3 Back to the original minimizer u , through the blow-up analysis, we prove that there is $u_0 \in K(\gamma, d)$ such that u_r and u_0 are sufficiently close in some sense. Intuitively, $\partial\{u > 0\} \cap B_r$ inherits properties of $\partial\{u_0 > 0\} \cap B_1$ for sufficiently small $r > 0$.

Generally, we will mainly consider Stage 1 and Stage 3. Stage 2 is complex, and we present some preliminary results.

2. BLOW-UP ANALYSIS

2.1. Initial regularity. Before we establish the blow-up analysis procedure, we should first consider the regularity theory for the minimizers of (1.1). Throughout this subsection, we assume that $u \in H_g^1(B_1)$ is a minimizer of (1.1) with $g \in \text{Lip}(\partial)$. With the help of the arguments in the proof of Theorem 1.2, we have the following result.

Lemma 2.1 (Boundedness). $0 \leq u \leq \sup_{\partial\Omega} g$.

Lemma 2.2 (Sub-harmonic function). u is sub-harmonic, i.e. $-\Delta u \leq 0$.

Proof. For $\varphi \in C_0^1(B_1)$ such that $\varphi \leq 0$, it follows from the minimizing property of u that

$$\int_{B_1} \left(\frac{1}{2} |\nabla u|^2 + u^\gamma \chi_{\{u>0\}} \right) \leq \int_{B_1} \left(\frac{1}{2} |\nabla(u + \varphi)|^2 + (u + \varphi)^\gamma \chi_{\{u+\varphi>0\}} \right). \quad (2.1)$$

Since φ is non-positive, we see that

$$\int_{B_1} u^\gamma \chi_{\{u>0\}} \geq \int_{B_1} \frac{1}{2} |\nabla(u + \varphi)|^2 + (u + \varphi)^\gamma \chi_{\{u+\varphi>0\}}.$$

Then, (2.1) implies that

$$\int_{B_1} |\nabla u|^2 \leq \int_{B_1} |\nabla(u + \varphi)|^2.$$

By the arbitrariness of φ , $-\Delta u \leq 0$ in the weak sense, that is, u is sub-harmonic. $\square$

The following proposition concerns the Hölder's continuity of u .

Proposition 2.3 (Hölder's continuity). $u \in C^{0,\alpha}(B_1)$ for any $\alpha \in (0, 1)$. In particular, for $R > 0$, $[u]_{C^{0,\alpha}(B_R)} \leq C$, where $C > 0$ depends only on $\alpha, d, \gamma, \|\nabla u\|_{L^2(B_1)}$, and $\|u\|_{L^\infty(B_1)}$.

The basic idea is that by Lemma 2.1, the boundedness of minimizers implies that the reaction term in the Alt-Phillips energy function is a small perturbation.

Proof. For simplicity, we assume that $\|u\|_{L^\infty(B_1)} + \|\nabla u\|_{L^2(B_1)} \leq C_0$. For $B_r(x_0) \subset B_1$, we define the harmonic replacement $H_{x_0,r}$ of u in $B_r(x_0)$ is the solution of the Dirichlet problem

$$\begin{cases} \Delta H_{x_0,r} = 0 & \text{in } B_r(x_0) \\ H_{x_0,r} = u & \text{on } \partial B_r(x_0). \end{cases}$$

Then, we can test the above equation by $u - H_{x_0,r}$ and obtain that

$$\begin{aligned} \int_{B_r(x_0)} |\nabla(u - H_{x_0,r})|^2 &= \int_{B_r(x_0)} (|\nabla u|^2 - |\nabla H_{x_0,r}|^2) - 2 \int_{B_r(x_0)} \nabla H_{x_0,r} \cdot \nabla(u - H_{x_0,r}) \\ &= \int_{B_r(x_0)} (|\nabla u|^2 - |\nabla H_{x_0,r}|^2). \end{aligned} \quad (2.2)$$

Recalling that u is a minimizer in B_1 , it follows that $E_\gamma(u, B_r(x_0)) \leq E_\gamma(H_{x_0,r}, B_r(x_0))$ and then

$$\begin{aligned} \int_{B_r(x_0)} |\nabla(u - H_{x_0,r})|^2 &= \int_{B_r(x_0)} (|\nabla u|^2 + 2u^\gamma \chi_{\{u>0\}} - |\nabla H_{x_0,r}|^2 - 2H_{x_0,r}^\gamma \chi_{\{H_{x_0,r}>0\}}) \\ &\quad + 2 \int_{B_r(x_0)} (H_{x_0,r}^\gamma \chi_{\{H_{x_0,r}>0\}} - u^\gamma \chi_{\{u>0\}}) \leq 2 \int_{B_r(x_0)} H_{x_0,r}^\gamma. \end{aligned} \quad (2.3)$$

Using the maximal principle, we have

$$\int_{B_r(x_0)} |\nabla(u - H_{x_0,r})|^2 \leq 2 \left(\sup_{B_1} u \right)^\gamma \mathcal{L}^d(B_r). \quad (2.4)$$

Let $0 < r < \rho$ and $H := H_{x_0,\rho}$ be the harmonic displacement of u in $B_\rho(x_0)$. Then,

$$\begin{aligned} \left(\int_{B_r(x_0)} |\nabla u|^2 \right)^{\frac{1}{2}} &\leq \left(\int_{B_r(x_0)} |\nabla(u - H)|^2 \right)^{\frac{1}{2}} + \left(\int_{B_r(x_0)} |\nabla H|^2 \right)^{\frac{1}{2}} \\ &\leq \left(\int_{B_\rho(x_0)} |\nabla(u - H)|^2 \right)^{\frac{1}{2}} + \left(\int_{B_r(x_0)} |\nabla H|^2 \right)^{\frac{1}{2}} \mathcal{L}^d(B_r)^{\frac{1}{2}} \\ &\leq C \mathcal{L}^d(B_\rho)^{\frac{1}{2}} + \left(\int_{B_r(x_0)} |\nabla H|^2 \right)^{\frac{1}{2}} \mathcal{L}^d(B_r)^{\frac{1}{2}}. \end{aligned} \quad (2.5)$$

Note that H is harmonic, we see that $|\nabla H|^2$ is sub-harmonic ($\Delta(|\nabla H|^2) \geq 0$) and then

$$0 < t \mapsto \int_{B_t(x_0)} |\nabla H|^2$$

is non-decreasing. As a result, we deduce from (2.5) that

$$\left(\int_{B_r(x_0)} |\nabla u|^2 \right)^{\frac{1}{2}} \leq C \rho^{\frac{d}{2}} + C \left(\int_{B_\rho(x_0)} |\nabla H|^2 \right)^{\frac{1}{2}} r^{\frac{d}{2}}.$$

Using (2.2) again, we have

$$\left(\int_{B_r(x_0)} |\nabla u|^2 \right)^{\frac{1}{2}} \leq C \rho^{\frac{d}{2}} + C \left(\int_{B_\rho(x_0)} |\nabla u|^2 \right)^{\frac{1}{2}} r^{\frac{d}{2}}. \quad (2.6)$$

Note that since H is harmonic, (2.6) is also a consequence that H minimizes the Dirichlet problem

$$\min \left\{ \int_{B_r(x_0)} |\nabla v|^2 : v = u \text{ on } \partial B_\rho(x_0) \right\}.$$

Define

$$D(x_0, r) := \left(\int_{B_r(x_0)} |\nabla u|^2 \right)^{\frac{1}{2}}.$$

It follows from (2.6) that

$$D(x_0, r) \leq C \left(\frac{\rho}{r} \right)^{\frac{d}{2}} + D(x_0, \rho),$$

where $C > 0$ depends only on C_0, d , and γ . We can apply standard iterative arguments that

$$D(x_0, r) \leq C(C_0, d, \gamma) \ln \left(\frac{1}{r} \right) + D \left(x_0, \frac{1}{2} \right),$$

where $0 < r < \frac{1}{2}$ and $x_0 \in B_{\frac{1}{2}}$. Thus, we have

$$\left(\int_{B_r(x_0)} |\nabla u|^2 \right)^{\frac{1}{2}} \leq C(\alpha, C_0, d, \gamma) r^{\alpha-1}.$$

Then, the Hölder's continuity directly follows from Morrey's lemma [2, Theorem 5.7]. $\square$

Remark 2.4. For a more general functional

$$E_{\gamma,\lambda}(u, \Omega) := \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 + \lambda u^{\gamma} \chi_{\{u>0\}} \right) dx.$$

With almost the same arguments, we can still obtain Lemma 2.1, 2.2, and Proposition 2.3. The only difference is that the constant $C > 0$ in Proposition 2.3 is also dependent on λ .

As a direct consequence, we have the following corollary.

Corollary 2.5. $\{u > 0\}$ is open.

Since $\{u > 0\}$ is open, we can perturb u in it and achieve the result below.

Lemma 2.6 (Euler-Lagrange equation). $\Delta u = \gamma u^{\gamma-1}$ in $\{u > 0\}$.

Proof. Let $\varphi \in C_0^\infty(B_1)$. For $t \in (-\varepsilon, \varepsilon)$ with $\varepsilon > 0$ being sufficiently small, $\{u + t\varphi > 0\} = \{u > 0\}$. As a result, we have

$$\left. \frac{d}{dt} \right|_{t=0} E_{\gamma}(u + t\varphi, B_1) = 0.$$

Note that we have an expansion that

$$\begin{aligned} E_{\gamma}(u + t\varphi, B_1) &= \int \frac{1}{2} |\nabla(u + t\varphi)|^2 + (u + t\varphi)^{\gamma} \chi_{\{u+t\varphi>0\}} \\ &= \int \frac{1}{2} |\nabla u|^2 + t \nabla u \cdot \nabla \varphi + u^{\gamma} + \gamma t u^{\gamma-1} \varphi + O(t^2). \end{aligned}$$

As a result, we see that in the distributional sense, $\Delta u = \gamma u^{\gamma-1}$ in $\{u > 0\}$. $\square$

By Lemma 2.1, for $\gamma \in [1, 2)$, we have a stronger property.

Lemma 2.7. If $\gamma \in [1, 2)$, then $\Delta u = \gamma u^{\gamma-1}$.

Proof. Since $\Delta u = \gamma u^{\gamma-1}$ in $\{u > 0\}$ and u is bounded, $u \in C^{1,\alpha}$ for any $\alpha \in (0, 1)$. Then, $\Delta u = \gamma u^{\gamma-1}$ follows from integration by parts. The key point is that ∇u also has α -order decay near the free boundary. $\square$

To close this subsection, we give the following comparison property.

Lemma 2.8 (Comparison). For $\gamma \in [1, 2)$ and $u, v \geq 0$ in B_1 . We assume that $\Delta u \geq \gamma u^{\gamma-1}$ and $\Delta v = \gamma v^{\gamma-1}$ in B_1 . If $u \leq v$ on ∂B_1 , then $u \leq v$ in B_1 .

Remark 2.9. Here, the super and sub-solutions are in the sense of viscosity solutions. For the definition and related materials, one can check [1, Section 1 & 4].

Proof. Without loss of generality, we assume that $u, v \in C^2$. Define

$$\bar{t} := \{t : u + t \leq v \text{ in } \overline{B_1}\}.$$

We need to show that $\bar{t} \geq 0$. If $\bar{t} < 0$, since $u \leq v$ on ∂B_1 , there is a contact point $x_0 \in B_1$ such that $u(x_0) + \bar{t} = v(x_0)$ and $u + \bar{t} \leq v$ in $\overline{B_1}$. Then, it follows from the assumption that

$$\gamma u^{\gamma-1}(x_0) \leq \Delta(u + \bar{t})(x_0) \leq \Delta v(x_0) \leq \gamma v^{\gamma-1}(x_0) = \gamma(u(x_0) + \bar{t})^{\gamma-1}.$$

Note that for the second inequality, we use the property of the estimate of the Hessian matrix at the minimal point. If $\gamma > 1$, it is a contradiction. If $\gamma = 1$, the result is trivial due to the comparison principle of sub-harmonic functions. $\square$

2.2. Scaling-invariant properties of a minimizer. We assume that u is a minimizer in B_1 .² As mentioned previously, for $0 \in \partial\{u > 0\}$ we want to examine the blow-up sequence

$$u_r(x) = r^{-\beta} u(rx), \quad x \in B_{r^{-1}}.$$

Using Lemma 2.6, we intend to obtain the index $\beta > 0$ such that u_r is also a solution for the same Euler-Lagrange equation. It follows from simple calculations that

$$\Delta u_r(x) = r^{2-\beta} \Delta u(rx) = r^{2-\beta} r^{\beta(\gamma-1)} u_r^\gamma(x).$$

As a result, we enforce that

$$2 - \beta + \beta(\gamma - 1) = 0 \implies \beta = \frac{2}{2 - \gamma}.$$

We call β as the scaling parameter. The following lemma is tricky and useful.

Lemma 2.10. *Let u be a minimizer in B_r . There exist $\bar{A}, c > 0$, depending only on d and γ ³ such that if $A \geq \bar{A}$ and $\int_{\partial B_1} u = Ar^\beta$, then $\inf_{B_{\frac{1}{2}}} u \geq cAr^\beta$.*

Before we prove this lemma, we first give some intuitions and applications of it.

Note that if u is a harmonic function, the property follows directly from Harnack's inequality. Let $u_A := A^{-1}u$, we see that

$$\Delta u_A = \gamma A^{\gamma-2} u_A^{\gamma-1}. \quad (2.7)$$

It corresponds to the energy functional

$$A^2 \int \left(\frac{|\nabla u_A|^2}{2} + A^{\gamma-2} u_A^\gamma \right). \quad (2.8)$$

When $A \rightarrow +\infty$, the diffusion term is in the dominant position, and the minimizer is more like a harmonic function.

As a direct consequence of Lemma 2.10, we have the following result.

Corollary 2.11. *Assume that u is a minimizer in B_1 , $x \in B_{\frac{1}{2}}$, and $r \in (0, \frac{1}{4})$. Then,*

$$\sup_{B_r(x)} u \leq C \left(\inf_{B_r(x)} u + r^\beta \right),$$

where $C > 0$ is a universal constant.

Proof. By Lemma 2.2, we see that u is sub-harmonic and then

$$\sup_{B_r(x)} u \leq C(d) \left(\int_{\partial B_{2r}(x)} u \right). \quad (2.9)$$

If $\int_{\partial B_{2r}(x)} u \leq \bar{A}r^\beta$, where $\bar{A}$ is as in Lemma 2.10, then (2.9) implies

$$\sup_{B_r(x)} u \leq \bar{A}r^\beta.$$

On the other hand, when $\int_{\partial B_{2r}(x)} u \geq \bar{A}r^\beta$, we apply Lemma 2.10 to deduce from (2.9) that

$$\inf_{B_r(x)} u \geq C^{-1}(d) \left(\int_{\partial B_{2r}(x)} u \right) \geq C^{-1}(d) \sup_{B_r(x)} u,$$

completing the proof. $\square$

Corollary 2.12. *If u is a minimizer in B_1 and $x \in \partial\{u > 0\} \cap B_{\frac{1}{2}}$ then for any $r \in (0, \frac{1}{2})$, $\sup_{B_r(x)} u \leq Cr^\beta$, where $C > 0$ is a universal constant.*

Proof. Since $0 \in \partial\{u > 0\}$, we have $\inf_{B_r(x)} u = 0$. The result follows from the application of Corollary 2.11. $\square$

²We say that u is a minimizer in $\Omega \subset \mathbb{R}^d$ if $u \in H_g^1(\Omega)$ minimizes (1.1) for some Lipschitz g .

³In the rest of this note, if a constant depends only on the numbers d and γ , then we call it as a universal constant.

Proof of Lemma 2.10. Define $c > 0$ as a constant that we will choose later. Assume that the result is not true, we have a sequence of minimizers in B_1 , given by $\{u_n\}$ such that $\int_{\partial B_1} u_n = A_n \rightarrow +\infty$ and $\inf_{B_{\frac{1}{2}}} u_n < cA_n$. Define $\bar{u}_n := A_n^{-1}u_n$. Then,

$$\int_{\partial B_1} \bar{u}_n = 1 \quad \text{and} \quad \inf_{B_{\frac{1}{2}}} \bar{u}_n < c. \quad (2.10)$$

Let H_n be the harmonic replacement of $\bar{u}_n$ in B_1 . By (2.10), we see that $\int_{\partial B_1} H_n = 1$. It follows from Harnack's inequality for harmonic functions that

$$\inf_{B_{\frac{1}{2}}} H_n \geq C(d). \quad (2.11)$$

By almost the same arguments in obtaining (2.3), we deduce from (2.8) that

$$\int_{B_1} |\nabla(\bar{u}_n - H)|^2 \leq 2A_n^{\gamma-2} \int_{B_1} H_n^\gamma. \quad (2.12)$$

If $\gamma \leq 1$, we use Hölder's inequality that

$$\int_{B_1} H_n^\gamma \leq \left(\int_{B_1} H_n \right)^\gamma \leq C \left(\int_{\partial B_1} H_n \right)^\gamma.$$

Here, for the last inequality, we have used the mean value formula of harmonic functions. Combining (2.12), we have

$$\int_{B_1} |\nabla(\bar{u}_n - H)|^2 \leq 2A_n^{\gamma-2} \rightarrow 0$$

as $n \rightarrow +\infty$. Using (2.10) again, we see that

$$0 \leq H_n \leq C(d) \text{ in } B_{\frac{1}{2}}.$$

Recalling that $\bar{u}_n$ is sub-harmonic (Lemma 2.2), the comparison principle yields that

$$0 \leq \bar{u}_n \leq H_n \leq C(d) \text{ in } B_{\frac{1}{2}}. \quad (2.13)$$

Moreover, (2.12) implies that

$$\sup_{n \in \mathbb{Z}_+} \int_{B_{\frac{1}{2}}} |\nabla \bar{u}_n|^2 \leq C.$$

This, together with the property that $\bar{u}_n$ minimizes (2.8), (2.13), and Remark 1.1 that $\|\bar{u}_n\|_{C^{0,\frac{1}{2}}(B_{\frac{1}{2}})} \leq C$ for any $n \in \mathbb{Z}_+$. Then, up to a subsequence, u_n and H_n both converge uniformly to a function u on $B_{\frac{1}{2}}$. As a result, it follows from (2.11) that

$$\inf_{B_{\frac{1}{2}}} \bar{u}_n \geq \frac{1}{2} \inf_{B_{\frac{1}{2}}} H_n \geq C_0(d).$$

Choosing sufficiently small $c > 0$, we obtain a contradiction.

Now, we assume that $\gamma > 1$. By Lemma 2.7, (2.7), we see that

$$\Delta \bar{u}_n = A_n^{\gamma-2} \gamma \bar{u}_n^{\gamma-1}.$$

Let $d_n := \bar{u}_n - H_n$. It satisfies the Dirichlet problem

$$\begin{cases} \Delta d_n = A_n^{\gamma-2} \gamma \bar{u}_n^{\gamma-1} & \text{in } B_1, \\ d_n = 0 & \text{in } \partial B_1. \end{cases}$$

For $\varphi \in C_0^1(B_1)$, we have

$$\int_{B_1} d_n \Delta \varphi = \int_{B_1} A_n^{\gamma-2} \gamma \bar{u}_n^{\gamma-1} \varphi.$$

It follows from (2.13)⁴

$$\left| \int_{B_1} A_n^{\gamma-2} \gamma \bar{u}_n^{\gamma-1} \varphi \right| \leq C A_n^{\gamma-2} \left(\int_{B_1} H_n \right)^{\gamma-1} \rightarrow 0.$$

Combining the boundary condition of d_n , we see that $d_n \rightarrow 0$ weakly in $L^p(B_1)$ for some $p > 1$. On the other hand, the convergence is uniform in $B_{\frac{1}{2}}$ due to (2.13) again. Then, the result follows from similar arguments in the first case. $\square$

Using Corollary 2.11 and 2.12, we deduce the following result on the Lipschitz estimate of minimizers.

Corollary 2.13. *Assume that u is a minimizer in B_1 and $\{u = 0\} \cap B_{\frac{1}{4}} = \emptyset$. Then, $|\nabla u|^2 \leq C u^\gamma$ in $B_{\frac{1}{2}}$, where $C > 0$ is a universal constant.*

Proof. We divide it into several cases, depending on the range of the index $\gamma \in [0, 2)$.

Case 1: $\gamma = 0$. Assume that $x \in \{u > 0\} \cap B_{\frac{1}{2}}$. Let $\ell := \text{dist}(x, \{u = 0\})$. It is easy to see that $\ell \in (0, \frac{1}{2}]$ and $B_\ell(x) \subset B_1$. Since $\gamma = 0$, it follows from Lemma 2.6 and Corollary 2.12 that

$$\begin{cases} \Delta u = 0 & \text{in } B_\ell(x), \\ 0 \leq u \leq C(d, \gamma)\ell & \text{in } B_\ell(x). \end{cases} \quad (2.14)$$

Then, we deduce that

$$|\nabla u(x)| \leq C(d, \gamma) \quad (2.15)$$

by using gradient estimates of harmonic functions. To close the proof with $\gamma = 0$, one needs to verify that u is Lipschitz in $B_{\frac{1}{2}}$. For $x, y \in B_{\frac{1}{2}}$, we consider the line $\overline{xy} \subset B_{\frac{1}{2}}$. Since $\{u > 0\}$ is open,

$$\overline{xy} \cap \{u > 0\} = \bigcup_{i \in \mathbb{Z}_+} \overline{a_i a_{i+1}},$$

where $u(a_i) = 0$. Moreover, we require that under a translation and scaling, $\{a_i\} \subset [0, 1]$ with $a_i < a_{i+1}$. By (2.15) and Lagrange mean value theorem, for any $i \in \mathbb{Z}_+$,

$$|u(a_i) - u(a_{i+1})| \leq C(d, \gamma)|a_i - a_{i+1}|.$$

Summing $i \in \mathbb{Z}_+$ up, we obtain that

$$|u(x) - u(y)| \leq \sum_{i \in \mathbb{Z}_+} |u(a_i) - u(a_{i+1})| \leq \sum_{i \in \mathbb{Z}_+} C|a_i - a_{i+1}| \leq C(d, \gamma)|x - y|.$$

Case 2: $\gamma \in (0, 2)$. In this case, we see that the scaling parameter $\beta = \frac{2}{2-\gamma} > 1$. It means that we can pointwise define the value of ∇u in the whole domain B_1 . In $\{u > 0\}$ and the interior of $\{u = 0\}$, ∇u trivially exists. For $x \in \partial\{u > 0\}$, it follows from Corollary 2.12 that

$$\frac{|u(x+h) - u(x)|}{h} \sim h^{\beta-1} \rightarrow 0^+$$

as $h \rightarrow 0^+$, which implies that $\nabla u(x) = 0$. When $\gamma \in (0, 1)$, $\Delta u = \gamma u^{\gamma-1}$ in $\{u > 0\}$. We cannot bound the right-hand side by simply using estimates as in (2.14). Extra tricks apply in this case. Fix $x \in B_{\frac{1}{2}}$. Define $\ell = \delta u^{\frac{1}{\beta}}(x)$, where $\delta \in (0, \frac{1}{10})$ is a constant to be determined. The estimate in Corollary 2.12 gives that

$$0 \leq u(x) \leq C \text{dist}(x, \{u = 0\})^\beta \leq C(d, \gamma).$$

If $\delta \ll 1$, then $B_\ell(x) \subset \{u > 0\} \cap B_1$. By Corollary 2.11, we obtain that

$$\sup_{B_\ell(x)} u \leq C(u(x) + \delta^\beta u(x)) \leq C(d, \gamma)u(x), \text{ if } \delta^\beta \leq 1 \quad (2.16)$$

and

$$u(x) \leq C(d, \delta) \left(\inf_{B_\ell(x)} u + \delta^\beta u(x) \right).$$

⁴Indeed, we use the property that $0 \leq \bar{u}_n \leq H_n$ in B_1 .

When $\delta = \delta(d, \gamma) > 0$ is sufficiently small, we combine (2.16) to get

$$\frac{\sup_{B_\ell(x)} u}{C} \leq u(x) \leq C \inf_{B_\ell(x)} u,$$

where $C > 0$ is a universal constant. In conclusion, we have

$$\begin{cases} \Delta u = \gamma u^{\gamma-1} & \text{in } B_\ell(x), \\ \frac{u(x)}{C(d, \gamma)} \leq u \leq C(d, \gamma)u(x) & \text{in } B_\ell(x). \end{cases}$$

Letting $u_\ell(y) := \ell^{-\beta} u(x + \ell y)$, we have

$$\begin{cases} \Delta u_\ell = \gamma u_\ell^{\gamma-1} & \text{in } B_\ell(x), \\ \frac{1}{C(d, \gamma)} \leq u_\ell \leq C(d, \gamma) & \text{in } B_\ell(x). \end{cases}$$

Using standard regularity theory of harmonic functions, we obtain that

$$|\nabla u_\ell(0)| \leq C(d, \gamma). \quad (2.17)$$

The desired property is a consequence of scaling back. $\square$

Remark 2.14. We can improve regularity estimates in (2.15) and 2.17 to every order k . The best interior estimates of minimizers in B_1 are determined by the decay rate near the free boundary $\partial\{u > 0\}$. For example, when $\gamma = 1$, we have $\beta = 2$ and $u \in C^{1,1}$.

Corollary 2.15. *Let u be a minimizer in B_1 , $0 \in \partial\{u > 0\}$, and $u_r(x) = r^{-\beta} u(rx)$ with $r > 0$. Then, for sufficiently small $r > 0$, $|\nabla u_r| \leq C$ in B_R , where $C > 0$ depends only on d, γ , and R .*

Proof. Fix $x \in B_R$. If $0 < r \ll 1$, then u_r is well defined in B_R . Using Corollary 2.13, we deduce that

$$|\nabla u_r(x)| \leq r^{1-\beta} |\nabla u(rx)| \leq \frac{Cu^{\frac{\beta-1}{\beta}}(rx)}{r^{\beta-1}} \leq \frac{C(|rx|^{\frac{\beta-1}{\beta}})^\beta}{r^{\beta-1}} \leq C(d, \gamma, R).$$

Here, for the last inequality, we have used Corollary 2.12. $\square$

The following lemma gives the non-degeneracy of minimizers of (1.1). It ensures that the blow-up limit is non-trivial.

Lemma 2.16. *Suppose that u is a minimizer in B_1 and $0 \in \overline{\{u > 0\}}$. Then,*

$$\sup_{\partial B_r} u \geq Cr^\beta,$$

where $C > 0$ is a universal constant. In particular, for $u_r(x) = r^{-\beta} u(x)$,

$$\sup_{\partial B_1} u_r \geq C.$$

Before we present the proof, we first consider its intuition. It is not hard to find that the non-degeneracy property holds only for minimizers. For example, when $\gamma = 0$, we see that minimizers enjoy $\Delta u = 0$ in $\{u > 0\}$. Consider a constant $c > 0$, it is obvious that $\Delta c = 0$ and one can choose $c > 0$ arbitrarily small, meaning that (2.16) is not true for merely the equation of the Euler-Lagrange equation. Such a contradiction occurs because if u is a minimizer, the small boundary condition enforces that u must contact 0 very fast, showing that 0 is an interior point of $\{u = 0\}$, which is impossible since we assume that $0 \in \partial\{u > 0\}$.

Lemma 2.16 is a direct result of the following auxiliary lemma.

Lemma 2.17. *Let u be a minimizer in B_1 . Then, there exists a universal constant $\bar{\varepsilon} > 0$ such that if $\sup_{\partial B_1} u < \bar{\varepsilon}$ then $u \equiv 0$ in $B_{\frac{1}{2}}$.*

Remark 2.18. The lemma above echoes the intuition that if the boundary condition is sufficiently small, then $0 \in B_1 \setminus \overline{\{u > 0\}}$.

Proof of Lemma 2.17. We prove by contradiction. Assume that $u \equiv 0$ is not true in $B_{\frac{1}{2}}$.

Case 1: Let $\gamma = 0$. We consider the solution of the Dirichlet problem

$$\begin{cases} \Delta \Phi = 0 & \text{in } B_1 \setminus B_{\frac{1}{2}}, \\ \Phi = 2\bar{\varepsilon} & \text{on } \partial B_1, \\ \Phi = 0 & \text{on } \partial B_{\frac{1}{2}}. \end{cases}$$

Indeed, we have

$$\Phi(x) = \frac{2\bar{\varepsilon} \left(\left(\frac{1}{2}\right)^{d-2} - \frac{1}{|x|^{d-2}} \right)}{\left(\frac{1}{2}\right)^{d-2} - 1}.$$

Let $v := \min\{u, \Phi\}$. Since u is a minimizer, we have the comparison that

$$E_0(u, B_1) \leq E_0(v, B_1).$$

Noting that $\Phi \equiv 0$ in $B_{\frac{1}{2}}$, it yields that $v(x) = 0$ for any $x \in B_{\frac{1}{2}}$. Then,

$$E_0(u, B_1) \leq E_0(v, B_1 \setminus B_{\frac{1}{2}}).$$

Particularly,

$$\int_{B_1} \frac{|\nabla u|^2}{2} + \chi_{\{u>0\}} \leq \int_{B_1 \setminus B_{\frac{1}{2}}} \frac{|\nabla v|^2}{2} + \chi_{\{v>0\}}.$$

As a result,

$$\int_{B_1 \setminus B_{\frac{1}{2}}} \left(\frac{|\nabla v|^2}{2} - \frac{|\nabla u|^2}{2} + \chi_{\{v>0\}} - \chi_{\{u>0\}} \right) \geq \int_{B_{\frac{1}{2}}} \left(\frac{|\nabla u|^2}{2} + \chi_{\{u>0\}} \right).$$

By the definition of v , $\chi_{\{v>0\}} - \chi_{\{u>0\}} \leq 0$, showing that

$$\int_{B_1 \setminus B_{\frac{1}{2}}} \left(\frac{|\nabla v|^2}{2} - \frac{|\nabla u|^2}{2} \right) \geq \int_{B_{\frac{1}{2}}} \left(\frac{|\nabla u|^2}{2} + \chi_{\{u>0\}} \right). \quad (2.18)$$

Using the trivial identity

$$|a|^2 - |b|^2 = -(a-b)^2 + 2a(a-b), \quad a, b \in \mathbb{R}^d,$$

with

$$a = \nabla v \quad \text{and} \quad b = \nabla u$$

in (2.18), we obtain that

$$\begin{aligned} \int_{B_{\frac{1}{2}}} \left(\frac{|\nabla u|^2}{2} + \chi_{\{u>0\}} \right) &\leq \int_{B_1 \setminus B_{\frac{1}{2}}} \left(-\frac{|\nabla(u-v)|^2}{2} + \nabla v \cdot \nabla(v-u) \right) \\ &\leq \int_{B_1 \setminus B_{\frac{1}{2}}} \nabla v \cdot \nabla(v-u). \end{aligned} \quad (2.19)$$

Integrating by parts, since $\Delta v = 0$ when $\gamma = 0$, we have

$$\begin{aligned} \int_{B_1 \setminus B_{\frac{1}{2}}} \nabla v \cdot \nabla(v-u) &= \int_{(B_1 \setminus B_{\frac{1}{2}}) \cap \{u>\Phi\}} \nabla v \cdot \nabla(v-u) \\ &= \int_{\partial(B_1 \setminus B_{\frac{1}{2}}) \cap \{u>\Phi\}} \partial_\nu v (v-u) \\ &= \int_{\partial(B_1 \setminus B_{\frac{1}{2}}) \cap \{u>\Phi\}} \partial_\nu \Phi (\Phi-u). \end{aligned} \quad (2.20)$$

On ∂B_1 , $\Phi = 2\bar{\varepsilon}$ and $u \leq \varepsilon$, implying that $v = u$. On $\partial\{\Phi > u\}$, we see that $\Phi = u$. Combining these basic ingredients, it follows that

$$\int_{B_1 \setminus B_{\frac{1}{2}}} \nabla v \cdot \nabla(v - u) = \int_{\partial B_{\frac{1}{2}}} (-u) \partial_\nu \Phi \leq C(d, \gamma) \bar{\varepsilon} \int_{\partial B_{\frac{1}{2}}} u, \quad (2.21)$$

where for the last inequality, we have used the explicit representation of Φ . This, together with (2.19) and (2.21), implies that

$$\int_{B_{\frac{1}{2}}} \left(\frac{|\nabla u|^2}{2} + \chi_{\{u>0\}} \right) \leq C(d, \gamma) \bar{\varepsilon} \left(\int_{\partial B_{\frac{1}{2}}} u \right). \quad (2.22)$$

By the trace theorem, we have

$$\int_{B_{\frac{1}{2}}} (|\nabla u| + u) \geq C(d, \gamma) \left(\int_{B_{\frac{1}{2}}} u \right). \quad (2.23)$$

For completeness, we sketch the proof of it. Without loss of generality, we consider the inequality on B_1 and ∂B_1 . By Newton-Leibniz formula, for any $x \in B_1$,

$$u\left(\frac{x}{|x|}\right) = u(x) + \int_{|x|}^1 \frac{\partial u}{\partial r} \left(t \frac{x}{|x|}\right) dt.$$

We integrate both sides of the above on B_1 and obtain (2.23). Since $\sup_{\partial B_1} u \leq \bar{\varepsilon} < \frac{1}{2}$, it follows from Lemma 2.1 that $0 \leq u \leq \bar{\varepsilon}$ in B_1 . Then, $\chi_{\{u>0\}} \geq u$. Moreover, we use Cauchy's inequality to arrive at

$$\int_{B_{\frac{1}{2}}} \left(\frac{|\nabla u|^2}{2} + \chi_{\{u>0\}} \right) \geq C \left(\int_{B_{\frac{1}{2}}} |\nabla u| \chi_{\{u>0\}} \right) \geq C \left(\int_{B_{\frac{1}{2}}} |\nabla u| \right).$$

Here, for the last inequality, we have used the property that $\nabla u = 0$ a.e. in the contact set. As a result,

$$\int_{B_{\frac{1}{2}}} \left(\frac{|\nabla u|^2}{2} + \chi_{\{u>0\}} \right) \geq C \int_{B_{\frac{1}{2}}} (|\nabla u| + u),$$

where $C > 0$ is an absolute constant. This, together with (2.23) and (2.22), implies that

$$\int_{\partial B_{\frac{1}{2}}} u \leq C \int_{B_{\frac{1}{2}}} \left(\frac{|\nabla u|^2}{2} + \chi_{\{u>0\}} \right) \leq C(d, \gamma) \bar{\varepsilon} \left(\int_{\partial B_{\frac{1}{2}}} u \right).$$

Since we assume that u is not identically equal to 0 in $B_{\frac{1}{2}}$, we can achieve the contradiction when $\bar{\varepsilon} = \bar{\varepsilon}(d, \gamma) > 0$ is sufficiently small.

Case 2: $\gamma \in (0, 1]$. Let Φ solve the following properties

$$\begin{cases} \Delta \Phi \leq \gamma \bar{\varepsilon}^{\gamma-1} & \text{in } B_1 \setminus B_{\frac{1}{2}}, \\ \Phi = 0 & \text{on } \partial B_{\frac{1}{2}}, \\ \Phi = 2\bar{\varepsilon} & \text{on } \partial B_1, \\ |\nabla \Phi| = 0 & \text{on } \partial B_{\frac{1}{2}}. \end{cases}$$

Indeed, we can choose Φ as

$$\Phi(x) = 4\bar{\varepsilon} \left(\left(|x| - \frac{1}{2} \right)_+ \right)^2.$$

Also let $v := \min\{u, \Phi\}$. With similar calculations in the first case, we obtain that

$$E_\gamma(v, B_1) \geq E_\gamma(u, B_1)$$

and then

$$\begin{aligned} \int_{B_{\frac{1}{2}}} \left(\frac{|\nabla u|^2}{2} + u^\gamma \right) &\leq \int_{B_1 \setminus B_{\frac{1}{2}}} \left(\frac{|\nabla v|^2}{2} - \frac{|\nabla u|^2}{2} + v^\gamma - u^\gamma \right) \\ &\leq \int_{B_1 \setminus B_{\frac{1}{2}}} (\nabla v \cdot \nabla(v - u) + v^\gamma - u^\gamma). \end{aligned} \quad (2.24)$$

Using integration by parts, and arguing as (2.21), it yields that

$$\begin{aligned} \int_{B_1 \setminus B_{\frac{1}{2}}} (\nabla v \cdot \nabla(v - u) + v^\gamma - u^\gamma) &= \int_{(B_1 \setminus B_{\frac{1}{2}}) \cap \{u > \Phi\}} (\nabla \Phi \cdot \nabla(\Phi - u) + \Phi^\gamma - u^\gamma) \\ &= \int_{(B_1 \setminus B_{\frac{1}{2}}) \cap \{u > \Phi\}} ((-\Delta \Phi)(\Phi - u) + (\Phi^\gamma - u^\gamma)) \\ &= \int_{(B_1 \setminus B_{\frac{1}{2}}) \cap \{u > \Phi\}} ((\Delta u - \Delta \Phi)(\Phi - u) + (\Phi^\gamma - u^\gamma) - \gamma u^{\gamma-1}(\Phi - u)). \end{aligned} \quad (2.25)$$

Since $\gamma \in (0, 1)$, we can use mean value theorem to get

$$(\Phi^\gamma - u^\gamma) - \gamma u^{\gamma-1}(\Phi - u) \leq 0$$

in $\{u > \Phi\}$. By Lemma 2.6 and the definition of Φ ,

$$\begin{aligned} \int_{(B_1 \setminus B_{\frac{1}{2}}) \cap \{u > \Phi\}} ((\Delta u - \Delta \Phi)(\Phi - u) + (\Phi^\gamma - u^\gamma) - \gamma u^{\gamma-1}(\Phi - u)) \\ \leq \int_{(B_1 \setminus B_{\frac{1}{2}}) \cap \{u > \Phi\}} (\gamma u^{\gamma-1} - \Delta \Phi)(\Phi - u) \leq 0. \end{aligned}$$

This, together with (2.24) and (2.25), gives a contradiction to the original assumption that $u \not\equiv 0$ in $B_{\frac{1}{2}}$.

Case 3: $\gamma \in (1, 2)$. Define $\Phi_x(y) := C_{d,\gamma}|y - x|^\beta$. We verify that $\Delta \Phi_x = \gamma \Phi_x^{\gamma-1}$. Choosing $\bar{\varepsilon} = \bar{\varepsilon}(d, \gamma) > 0$ sufficiently small, we see that for any $x \in B_{\frac{1}{2}}$, $\Phi_x \geq u$ on ∂B_1 . By Lemma 2.8, we see that $0 \leq u(x) \leq \Phi_x(x) = 0$. As a result, $u \equiv 0$ in $B_{\frac{1}{2}}$. $\square$

Remark 2.19. For $\gamma \in [1, 2)$, we do not use the minimizing property, since when $\gamma \geq 1$, the energy functional is convex and the solution of the Euler-Lagrange equation is almost equivalent to the minimizer.

Combining Lemma 2.12 and 2.16, we summarize the following result.

Proposition 2.20. *Let u be a minimizer in B_1 . Assume that $0 \in \partial\{u > 0\}$. Then*

$$\frac{r^\beta}{C} \leq \sup_{\partial B_r(x)} u \leq \sup_{B_r(x)} u \leq C r^\beta,$$

where $C > 0$ is a universal constant.

2.3. Compactness. We now consider the blow-up limits of minimizers.

Proposition 2.21. *Let u be a minimizer. For any $r > 0$, define the Weiss density as*

$$W(u, r) := E_\gamma(u_r, B_1) - \frac{\beta}{2} \int_{\partial B_1} u_r^2,$$

where we recall that $u_r(x) = r^{-\beta} u(rx)$. Then,

$$\frac{d}{dr} W(u, r) \geq \frac{1}{2r} \int_{\partial B_r} \left(\frac{\partial u_r}{\partial \nu} - \beta u_r \right)^2. \quad (2.26)$$

Proof. For $t \in (0, 1]$. Let

$$v_t(x) := \begin{cases} |x|^\beta u\left(\frac{x}{|x|}\right) & \text{if } x \in B_1 \setminus B_t, \\ t^\beta u\left(\frac{x}{t}\right) & \text{if } x \in B_t. \end{cases}$$

By direct computations, we have

$$E_\gamma(v_t, B_t) = \int_{B_t} \left(\frac{|\nabla v_t|^2}{2} + v_t^\gamma \right) = t^{d+2(\beta-1)} \int_{B_1} \left(\frac{|\nabla u|^2}{2} + u^\gamma \right),$$

and

$$E_\gamma(v_t, B_1 \setminus B_t) = \int_t^1 \left(\int_{\partial B_r} \frac{\beta^2 r^{2(\beta-1)} u^2\left(\frac{x}{|x|}\right)}{2} + \frac{r^{2(\beta-1)} \left| \partial_\tau u\left(\frac{x}{|x|}\right) \right|^2}{2} + r^{\beta\gamma} u^\gamma\left(\frac{x}{|x|}\right) \right),$$

where we have used

$$\nabla v_t(x) = (\partial_\nu v_t(x), \partial_\tau v_t(x)) = \left(\beta |x|^{\beta-1} u\left(\frac{x}{|x|}\right), |x|^{\beta-1} \nabla_\tau u\left(\frac{x}{|x|}\right) \right).$$

Here, we use the notation that

∂_ν : the normal derivative,

∂_τ : the tangent derivative.

Since u is a minimizer, we deduce that

$$E_\gamma(v_t, B_1) \geq E_\gamma(u, B_1).$$

Note that $v_1 = u$, we have

$$\frac{d}{dt} \Big|_{t=1} E_\gamma(v_t, B_1) \leq 0,$$

implying that

$$(d + 2(\beta - 1))E_\gamma(u, B_1) - \int_{\partial B_1} \left(\frac{\beta^2 u^2}{2} + \frac{|\partial_\tau u|^2}{2} + u^\gamma \right) \leq 0.$$

As a result,

$$\begin{aligned} \frac{d}{dr} \Big|_{r=1} E_\gamma(u_r, B_1) &= -(d + 2(\beta - 1))E_\gamma(u, B_1) + \int_{\partial B_1} \left(\frac{|\nabla u|^2}{2} + u^\gamma \right) \\ &\geq \int_{\partial B_1} \left(\frac{|\nabla u|^2}{2} + u^\gamma \right) - \int_{\partial B_1} \left(\frac{\beta^2 u^2}{2} + \frac{|\partial_\tau u|^2}{2} + u^\gamma \right) \\ &= \int_{\partial B_1} \left(\frac{1}{2} \left(\frac{\partial u}{\partial \nu} \right)^2 - \frac{\beta^2 u^2}{2} \right). \end{aligned}$$

Then,

$$\begin{aligned} \frac{d}{dr} \Big|_{r=1} E_\gamma(u_r, B_1) &\geq \int_{\partial B_1} \left(\frac{1}{2} \left(\frac{\partial u}{\partial \nu} \right)^2 - \frac{\beta^2 u^2}{2} \right) \\ &= \frac{1}{2} \int_{\partial B_1} \left(\frac{\partial u}{\partial \nu} - \beta u \right)^2 + 2\beta u \left(\frac{\partial u}{\partial \nu} - \beta u \right). \end{aligned} \tag{2.27}$$

Note that

$$\frac{d}{dr} \Big|_{r=1} u_r^2(x) = 2u \left(\frac{\partial u}{\partial \nu} - \beta u \right).$$

Combined with (2.27), it follows that

$$\frac{d}{dr} \Big|_{r=1} W(u, r) = \frac{d}{dr} \Big|_{r=1} \left(E_\gamma(u_r, B_1) - \beta \int_{\partial B_1} u_r^2 \right) \geq \frac{1}{2} \int_{\partial B_1} \left(\frac{\partial u}{\partial \nu} - \beta u \right)^2.$$

Since

$$W(u, rR) = W(u_r, R) = W(u_{rR}, 0),$$

we have

$$\begin{aligned} \left. \frac{d}{dr} \right|_{r=s} &= \lim_{h \rightarrow 0} \frac{W(u, s+h) - W(u, s)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{W(u_s, 1 + \frac{h}{s}) - W(u_s, 1)}{\frac{h}{s}} \cdot \frac{1}{s} \\ &\geq \frac{1}{2s} \int_{\partial B_1} \left(\frac{\partial u_s}{\partial \nu} - \beta u_s \right)^2, \end{aligned}$$

completing the proof. $\square$

With the help of Proposition 2.21 and results in previous subsections, we obtain the following properties on the limits of blow-up sequences.

Theorem 2.22 (Compactness and homogeneity). *Assume that u is a minimizer in B_1 . Let $0 \in \partial\{u > 0\}$. For $u_r(x) = r^{-\beta}u(rx)$. There exist $r_n \rightarrow 0^+$ such that $u_{r_n} \rightarrow u_0$ strongly in $H_{\text{loc}}^1(\mathbb{R}^d)$ and $C_{\text{loc}}^{0,\alpha}(\mathbb{R}^d)$ for any $\alpha \in (0, 1)$. Moreover, u_0 is β -homogeneous minimizer in $\mathbb{R}^d$ and $0 \in \partial\{u_0 > 0\}$.*

Proof. Corollary 2.15 ensures the weak convergence in $H_{\text{loc}}^1(\mathbb{R}^d)$ and strong convergence in $C^{0,\alpha}(\mathbb{R}^d)$ up to a subsequence. The β -homogeneity of u_0 follows from Proposition 2.21 if the strong convergence in $H_{\text{loc}}^1(\mathbb{R}^d)$ holds. Indeed, by Corollary 2.12, $W(u, r)$ is bounded, and $\lim_{r \rightarrow 0^+} W(u, r)$ exists due to the monotonicity. Then,

$$W(u_0, R) = \lim_{n \rightarrow +\infty} W(u_{r_n}, R) = \lim_{n \rightarrow +\infty} W(u, r_n R) = \lim_{r \rightarrow 0} W(u, r).$$

As a result, (2.26) implies that for

$$x \cdot \nabla u = \beta u \text{ in } \mathbb{R}^d,$$

showing that u is β -homogeneous.

Since $0 \in \partial\{u > 0\}$, we have $u(0) = 0$. The $C_{\text{loc}}^{0,\alpha}(\mathbb{R}^d)$ convergence shows that $u_0(0) = 0$. If $0 \notin \partial\{u_0 > 0\}$, then there is $r_0 > 0$ such that $u(B_{r_0}(x)) \equiv 0$, contradicting to Proposition 2.20. Finally, we show the strong convergence in $H_{\text{loc}}^1(\mathbb{R}^d)$ and the property that u_0 is a minimizer in $\mathbb{R}^d$. Let $u_n := u_{r_n}$ and choose $\varepsilon \in (0, \frac{1}{2})$. Find v_n such that

$$\begin{cases} \Delta v_n = 0 & \text{in } B_1 \setminus B_{1-\varepsilon}, \\ v_n = u_n & \text{on } \partial B_1, \\ v_n = u_0 & \text{on } \partial B_{1-\varepsilon}. \end{cases}$$

By Corollary 2.12, 2.13 and regularity estimate of harmonic functions, we have

$$|v_n| + |\nabla v_n| \leq C(d, \gamma) \text{ in } B_1 \setminus B_{1-\varepsilon}. \quad (2.28)$$

Here, we need the Lipschitz bound of u_0 . To get it, we further assume that $\nabla u_n \rightharpoonup^* \nabla u_0$ in L_{loc}^∞ . Define

$$w_n(x) := \begin{cases} u_0(x) & \text{if } x \in B_{1-\varepsilon}, \\ v_n(x) & \text{if } x \in B_1 \setminus B_{1-\varepsilon}. \end{cases}$$

Since u is a minimizer, it follows from (2.28) that

$$E_\gamma(u_n, B_1) \leq E_\gamma(w_n, B_1) = E_\gamma(u_0, B_1) + o_{n \rightarrow +\infty}(1).$$

On the other hand, the weak convergence in H^1 implies that

$$E_\gamma(u_0, B_1) \leq \liminf_{n \rightarrow +\infty} E_\gamma(u_n, B_1),$$

showing that $u_n \rightarrow u_0$ strongly in $H^1(B_1)$. With similar arguments, the convergence is strong in $H_{\text{loc}}^1(\mathbb{R}^d)$. For any v such that $v = u_0$ on ∂B_1 , we change the boundary value of v_n at $\partial B_{1-\varepsilon}$ by v and define

$$w_n(x) := \begin{cases} v(x) & \text{if } x \in B_{1-\varepsilon}, \\ v_n(x) & \text{if } x \in B_1 \setminus B_{1-\varepsilon}. \end{cases}$$

Again, by the minimality of u_n , we have

$$E_\gamma(u_n, B_1) \leq E_\gamma(w_n, B_1) = E_\gamma(v, B_1) + o_{n \rightarrow +\infty}(1).$$

By the strong convergence, we see that

$$E_\gamma(u_0, B_1) \leq E_\gamma(v, B_1).$$

Extended arguments show that u_0 is a minimizer. $\square$

3. IMPROVEMENT OF FLATNESS

Theorem 3.1. *Assume that u is a minimizer in B_1 . Let*

$$u_{\text{flat}} = C_\gamma[(x_1)_+]^\beta, \quad (3.1)$$

with

$$C_\gamma = \left(\frac{2}{\beta^2} \right)^{\frac{1}{2-\gamma}}. \quad (3.2)$$

Then, there exists a universal constant $\bar{\varepsilon} > 0$ such that if

$$\|u - u_{\text{flat}}\|_{L^\infty(B_1)} < \bar{\varepsilon},$$

then $\partial\{u > 0\}$ is a smooth hypersurface in $B_{\frac{1}{2}}$.

Intuitively, the above theorem is another form of ε -regularity of the free boundary. It implies that the smoothness of the free boundary follows from the closeness with the flat solution.

In $\mathbb{R}^2$ with $\gamma = 1$ consider the example $p = \frac{1}{2}(x^2 + y^2)$. We see that the contact set of p is the isolate point $(0, 0)$. Note that p does not satisfies the assumption of Theorem 3.1.

Before we talk about the proof of Theorem 3.1, we first give a couple of direct consequences as applications.

Corollary 3.2. *Let u be a minimizer in B_1 and $0 \in \partial\{u > 0\}$. If there exists $r_n \rightarrow 0^+$ such that*

$$u_{r_n} = r_n^{-\beta} u(r_n \cdot) \rightarrow C_\gamma[(x_1)_+]^\beta$$

as in Theorem 2.22, then $\partial\{u > 0\}$ is a smooth hypersurface in B_s .

Proof. Since the smoothness of $\{u > 0\}$ will not change under scaling, the result follows from the application of Theorem 3.1 to u_{r_n} for sufficiently small $r_n > 0$. $\square$

As a result, we can use the blow-up analysis in the previous section to characterize the regularity of free boundary at a specific point.

Corollary 3.3. *If the only minimal cone in $K(\gamma, d)$ is $C_\gamma[(x_1)_+]^2$, then $\partial\{u > 0\}$ is a smooth hypersurface in B_1 .*

Proof. By Theorem 3.1 and Corollary 3.2, $\partial\{u > 0\}$ is a smooth hypersurface in B_1 . $\square$

3.1. Flat cone. Since the assumption of Theorem 3.1 requires that the minimizer is approximated by a flat cone as given by (3.1). In this subsection, we explore the minimality of flat cone.

We first consider the minimal cone in $\mathbb{R}$.

Case 1: $\gamma = 0$. On $(-1, 1)$, Lemma 2.6 implies that $u'' = 0$ in $\partial\{u > 0\}$. Since u is a cone, we have that $u(0) = 0$ and $u(\lambda x) = \lambda u(x)$ for any $\lambda > 0$ and $x \in (-1, 1)$. If $u(-1) = a, u(1) = b > 0$, then

$$u(x) = \begin{cases} -ax & \text{if } x \in [-1, 0), \\ bx & \text{if } x \in [0, 1]. \end{cases}$$

We claim that such u cannot be a minimizer. It is because the linear function whose figure connects points $(-1, a)$ and $(1, b)$ has small energy $E_0(u, (-1, 1))$. As a result, a minimal cone must be in the form

$$u(x) = \begin{cases} 0 & \text{if } x \in [-1, 0), \\ u(1)x & \text{if } x \in [0, 1]. \end{cases}$$

By Proposition 2.20,

$$\frac{r}{C(d, \gamma)} \leq \sup_{(-r, r)} u \leq C(d, \gamma)r,$$

which implies that $u(1)$ has a universal upper and lower bound. However, from the Euler-Lagrange equation, for different $u(1)$ in the desired range from the estimates of the growth, we do not know which one we should choose. To this end, we adopt the minimality of u and construct the comparison function. Define $\{\ell_h\}_{h \in (-1, 1)}$ as the step function by

$$\ell_h(x) := \begin{cases} 0 & \text{if } x \in [-1, h), \\ \frac{x-h}{1-h} & \text{if } x \in [h, 1]. \end{cases}$$

Let $u_h(x) := u(\ell_h(x))$ be the comparison function. We see that $u'_h(x) = \frac{u'(\ell_h(x))}{1-h}$ and then

$$E_0(u_h, (-1, 1)) = \int_0^1 \frac{(u')^2}{2(1-h)} + (1-h).$$

Consequently,

$$\frac{d}{dh} \Big|_{h=0} E_0(u_h, (-1, 1)) = \frac{1}{2} \int_0^1 (u')^2 - 1 = 0.$$

Thus, $u(1) = \sqrt{2}$. It implies that the only potential minimal cone in $(-1, 1)$ is

$$u(x) = \begin{cases} 0 & \text{if } x \in [-1, 0), \\ \sqrt{2}x & \text{if } x \in [0, 1]. \end{cases}$$

On the other hand, we see that $K(1, 0)$ is not empty. Indeed, if we set the sufficiently small boundary condition on -1 and 1 , Lemma 2.17 shows that u and $\partial\{u > 0\}$ are both non-trivial. Conducting blow-up at points in $\partial\{u > 0\}$, we must obtain at least one element in $K(1, 0)$.

Case 2: $\gamma \in (0, 2)$. We see that $u'' = \gamma u^{\gamma-1}$. If $u \in K(1, \gamma)$, then $u(x) = u(1)x^\beta$ on $(0, 1)$. Consequently,

$$\beta(\beta - 1)u(1) = \gamma u^{\gamma-1}(1).$$

Then $u(1) = C_\gamma$ as in (3.2). In conclusion, the potential minimal cone are

$$u^{(1)} := \begin{cases} 0 & \text{if } x \in [-1, 0), \\ C_\gamma x^\beta & \text{if } x \in [0, 1], \end{cases} \quad \text{and} \quad u^{(2)} := \begin{cases} C_\gamma (-x)^\beta & \text{if } x \in [-1, 0), \\ C_\gamma x^\beta & \text{if } x \in [0, 1]. \end{cases}$$

For $\gamma \in [1, 2)$, since E_γ is convex, $u^{(1)}$ and $u^{(2)}$ are both minimal cone. For $\gamma \in (0, 1)$, $u^{(1)}$ is a minimal cone since we can repeat the blow-up arguments in the first case. We now prove that $u = u^{(2)}$ is not a minimal cone. We construct a comparison function that has smaller energy. Define $\varphi_0 \in C_0^\infty((-1, 0))$ such that

$$\varphi_0 \equiv 1 \text{ in } \left[-\frac{1}{2}, 0\right], \quad \varphi \text{ is non-decreasing in } \left[-1, -\frac{1}{2}\right].$$

Let

$$\varphi(x) := \begin{cases} \varphi_0(x) & \text{if } x \in [-1, 0], \\ \varphi_0(-x) & \text{if } x \in [0, 1]. \end{cases}$$

Through simple calculations,

$$\begin{aligned} E_\gamma(u + t\varphi, (-1, 1)) &= \int \frac{|\nabla u|^2 + 2t(\nabla u \cdot \nabla \varphi) + t^2|\nabla \varphi|^2}{2} + (u + t\varphi)^\gamma \\ &= \int \left(\frac{|\nabla u|^2}{2} + u^\gamma \right) + \int (t(\nabla u \cdot \nabla \varphi) + (u + t\varphi)^\gamma - u^\gamma) + O(t^2) \\ &= E_\gamma(u, (-1, 1)) + \int (-tu''\varphi + (u + t\varphi)^\gamma - u^\gamma) + O(t^2), \end{aligned}$$

where for the last equality, we have use integration by parts and the definition of φ , together with the property that $u(1) = u(-1)$. Using Lemma 2.6, we have

$$E_\gamma(u + t\varphi, (-1, 1)) = E_\gamma(u, (-1, 1)) + \int (-\gamma tu^{\gamma-1}\varphi + (u + t\varphi)^\gamma - u^\gamma) + O(t^2). \quad (3.3)$$

When $\gamma \in (0, 1)$ the function $f(t) = t^\gamma$ is concave. As a result,

$$(-\gamma tu^{\gamma-1}\varphi + (u + t\varphi)^\gamma - u^\gamma) \leq 0. \quad (3.4)$$

By Taylor expansion,

$$f(u + t\varphi) - (f(u) + f'(u)t\varphi) = \frac{1}{2}f''(\xi)(t\varphi)^2,$$

where $\xi \in (u, u + t\varphi)$. (3.4) yields that

$$\begin{aligned} \int (-\gamma tu^{\gamma-1}\varphi + (u + t\varphi)^\gamma - u^\gamma) &\leq \int_0^{t^{\frac{1}{\beta}}} (-\gamma tu^{\gamma-1}\varphi + (u + t\varphi)^\gamma - u^\gamma) \\ &\leq -C \int_0^{t^{\frac{1}{\beta}}} t^\gamma \varphi^2 \leq -C(d, \gamma)t^{\gamma+\frac{1}{\beta}}. \end{aligned} \quad (3.5)$$

Since $\beta = \frac{2}{2-\gamma}$, we have

$$\gamma + \frac{1}{\beta} = 1 + \frac{\gamma}{2} < 2,$$

as $\gamma \in (0, 1)$. Combining (3.3) and (3.5), if $t > 0$ is sufficiently small, then

$$E_\gamma(u + t\varphi, (-1, 1)) < E_\gamma(u, (-1, 1)),$$

showing that u is not a minimizer.

We summarize the above analysis into the following proposition.

Proposition 3.4. *In $\mathbb{R}$, if u is a minimizer in $(-1, 1)$, then the following properties hold.*

(1) *If $\gamma = 0$, the minimal cone is*

$$u(x) = \begin{cases} 0 & \text{if } x \in [-1, 0], \\ \sqrt{2}x & \text{if } x \in [0, 1]. \end{cases}$$

(2) *If $\gamma \in (0, 1)$, the minimal cone is*

$$u = \begin{cases} 0 & \text{if } x \in [-1, 0], \\ C_\gamma x^\beta & \text{if } x \in [0, 1]. \end{cases}$$

(3) *If $\gamma \in [1, 2)$, there are two different minimal cones*

$$u^{(1)} = \begin{cases} 0 & \text{if } x \in [-1, 0], \\ C_\gamma x^\beta & \text{if } x \in [0, 1], \end{cases} \quad \text{and} \quad u^{(2)} := \begin{cases} C_\gamma (-x)^\beta & \text{if } x \in [-1, 0], \\ C_\gamma x^\beta & \text{if } x \in [0, 1]. \end{cases}$$

To construct minimal cones in $\mathbb{R}^d$, we can trivialize a couple of variables. In particular, we need the following result.

Proposition 3.5. *Assume that $U : \mathbb{R}^d \rightarrow \mathbb{R}$ and $U(x_1, x_2, \dots, x_d) = u(x_1, x_2, \dots, x_{d-1})$. Then, U is a minimizer in $\mathbb{R}^d$ if and only if u is a minimizer in $\mathbb{R}^{d-1}$.*

Proof. First we assume that u is a minimizer in $\mathbb{R}^d$. If U is not a minimizer in $\mathbb{R}^d$, then, up to a translation, there must be some $R > 0$ and $v : B_R^{d-1} \times [-R, R]$ (here B_R^{d-1} represents the $(d-1)$ -ball in $\mathbb{R}^{d-1}$) such that $v = U$ on $\partial B_R^{d-1} \times [-R, R]$ and

$$E_\gamma(v, B_R^{d-1} \times [-R, R]) \leq E_\gamma(U, B_R^{d-1} \times [-R, R]).$$

Consequently,

$$\int_{-R}^R \int_{B_R^{d-1}} \left(\frac{|\nabla v|^2}{2} + v^\gamma \right) dx' dx_d < \int_{-R}^R \int_{B_R^{d-1}} \left(\frac{|\nabla U|^2}{2} + U^\gamma \right) dx' dx_d.$$

By average arguments, there must be some $t \in (-R, R)$ such that

$$E_\gamma(v(\cdot, t), B_R^{d-1}) < E_\gamma(u, B_R^{d-1}),$$

a contradiction to the minimality of u .

On the other hand, we assume that u is not a minimizer and U is a minimizer. Up to a translation, there is B_R^{d-1} and $v : B_R^{d-1} \rightarrow \mathbb{R}$ such that $v = u$ on ∂B_R^{d-1} , and

$$E_\gamma(v, B_R^{d-1}) = E_\gamma(u, B_R^{d-1}) - \delta, \quad (3.6)$$

where $\delta > 0$. Let $L > 0$. We define a non-negative smooth function g such that $g \equiv 1$ in $[-L, L]$ and $g \equiv 0$ in $\mathbb{R} \setminus [-L-1, L+1]$. Define

$$V := g(x_d)v + (1 - g(x_d))u.$$

Then, V is an admissible comparison of U . Through simple calculations, we see that

$$E_\gamma(V, B_R^{d-1} \times [-L-1, L+1]) \leq E_\gamma(V, B_R^{d-1} \times [-L, L]) + M,$$

where $M > 0$ is independent of L . By (3.6) and the assumption of this proposition, we have

$$E_\gamma(V, B_R^{d-1} \times [-L-1, L+1]) \leq E_\gamma(U, B_R^{d-1} \times [-L-1, L+1]) - 2L\delta + M.$$

Letting $L \rightarrow +\infty$, it contradicts to the minimality of U . $\square$

3.2. Transformed minimizers. Assume that u is a minimizer in B_1 , we define the transform of this minimizer as

$$v := \frac{\beta}{\sqrt{2}} u^{\frac{1}{\beta}}.$$

Let

$$v_{\text{flat}} := \frac{\beta}{\sqrt{2}} u_{\text{flat}}^{\frac{1}{\beta}} = (x_1)_+.$$

We now transform the problem on u to v . After this transform, v satisfies the system below.

$$\begin{cases} \Delta v = \frac{(\beta-1)(1-|\nabla v|^2)}{v}, & \text{in } \partial\{u > 0\}, \\ |v - v_{\text{flat}}| < \varepsilon < \bar{\varepsilon} & \text{in } B_1, \\ |\nabla v| = 1 & \text{on } \partial\{u > 0\}. \end{cases} \quad (3.7)$$

Note that ε and $\bar{\varepsilon}$ may not be the original ones, possibly with some exponential of multiplicity indexes. The condition $|\nabla v| = 1$ is in viscosity sense. We present the definition of viscosity solution of $|\nabla v(0)| \geq 1$ with $0 \in \partial\{v > 0\}$ for example. It implies that for any $\varphi \in C^\infty$, $\varphi \geq u$ in B_δ for some $\delta > 0$ with $\varphi(0) = u(0)$ and $|\nabla \varphi(0)| \neq 0$, there must be $|\nabla \varphi(0)| \geq 1$. This condition on the free boundary arise from the inner perturbation of the minimizer v . One can refer to the arguments in [3, Page 4] for similar arguments.

The following lemma is crucial in the proof of the improvement of flatness.

Lemma 3.6. *Assume that v solves (3.7). Suppose that $0 \in \partial\{v > 0\}$ and*

$$(x_1 - \varepsilon)_+ \leq v \leq (x_1 + \varepsilon)_+ \text{ in } B_1.$$

There are $r_0 \in (0, 1)$ and $\bar{\varepsilon} > 0$, being two universal constants such that if $0 < \varepsilon < \bar{\varepsilon}$, there is $e \in \mathbb{S}^{n-1}$ such that

$$\left(x \cdot e - \frac{1}{2}\varepsilon r_0\right)_+ \leq v \leq \left(x \cdot e + \frac{1}{2}r_0\varepsilon\right)_+ \text{ in } B_{r_0},$$

and

$$|e - e_1| \leq C\varepsilon, \tag{3.8}$$

where $C > 0$ is a universal constant.

Before we give the proof of Lemma 3.6, we first assume the validness of it and provide the application.

If we conduct the scaling $v_{r_0}(x) = r_0^{-1}v(r_0x)$ and obtain that

$$\left(x \cdot e - \frac{1}{2}\varepsilon\right)_+ \leq v \leq \left(x \cdot e + \frac{1}{2}\varepsilon\right)_+ \text{ in } B_1.$$

By applying Lemma 3.6 repeatedly, we have

$$\left(x \cdot e^k - \frac{\varepsilon}{2^k}\right)_+ \leq v_{r_0^k} \leq \left(x \cdot e^k + \frac{\varepsilon}{2^k}\right)_+.$$

As a result, taking limit of $e^k \rightarrow e^\infty$, it follows that

$$v \sim (x \cdot e^\infty \pm \varepsilon r^{1+\alpha}) \text{ in } B_r, \tag{3.9}$$

where we let $\alpha = \log_{r_0} \frac{1}{2}$. Note that by (3.8), the convergence of e^k enjoys the estimate that

$$|e^k - e^\infty| \leq \frac{\varepsilon}{2^k}.$$

Consequently, (3.9) implies that

$$\partial\{v > 0\} \cap B_r \subset B_{r^{1+\alpha}}(H_{e^\infty}) \cap B_r,$$

where H_{e^∞} is the hyperplane whose normal vector is e^∞ . By the Sandwich theorem, we see that $\partial\{v > 0\}$ is $C^{1,\alpha}$ at the point 0. For some point p near 0, such as $y \in \partial\{v > 0\} \cap B_{\varepsilon^2}$. We see that

$$w(x + y) \sim (x_1 \pm (|y_1| + \varepsilon))_+.$$

By applying Lemma 2.10 again at the point y , same regularity properties hold. Recovering u from the tranformed function v , we conclude Theorem 3.1.

We now prove Lemma 2.10. To make it clear, we outline the idea of the proof. Consider a nonlinear system $F(u) = 0$ with $|u - p| < \varepsilon$ in B_1 , where $p \in P$ with P being a very nice collection such that $F(P) \equiv 0$. From the statement of Lemma 2.10, we intend to show that there exists $p' \in P$ such that

$$|u - p'| < \frac{1}{2}\varepsilon r_0^\lambda \text{ in } B_{r_0},$$

where r_0 and λ are constants to be chosen. We consider the perturbation

$$u_\varepsilon = p + \varepsilon \hat{u}_\varepsilon.$$

Then,

$$0 = \frac{F(p + \varepsilon \hat{u}_\varepsilon) - F(p)}{\varepsilon}.$$

Letting $\varepsilon \rightarrow 0^+$, assume that $\hat{u}_\varepsilon \rightarrow \hat{u}_0$, we have

$$F'(p)\tilde{u}_0 = 0.$$

We now set $\hat{u}_0 = q + r_0^{\lambda+1}$ in B_{r_0} . It follows that

$$u_\varepsilon = p + \varepsilon \hat{u}_\varepsilon = p + \varepsilon \hat{u}_0 + \varepsilon o(1). \quad = p + \varepsilon q + \varepsilon r^{\lambda+1} + \varepsilon o(1).$$

Choosing sufficiently small ε , we have

$$u_\varepsilon = p + \varepsilon q + \frac{\varepsilon}{50} r_0^\lambda \text{ in } B_{r_0}.$$

Adjust $p + \varepsilon q$ to some $p' \in P$, we obtain the proof.

Proof of Lemma 2.10. We consider the inner perturbation

$$u_\varepsilon = (x_1 + \varepsilon w_\varepsilon)_+.$$

Since we require that u_ε solves (3.7), w_ε satisfies the system

$$\begin{cases} \Delta w = \frac{(\beta-1)}{x+\varepsilon w} \left(-2 \frac{\partial w}{\partial x_1} - \varepsilon |\nabla w|^2 \right) & \text{in } \{u > 0\}. \\ \frac{\partial w}{\partial x_1} = O(\varepsilon) & \text{on } \partial\{u > 0\} \sim \{x_1 = 0\} \\ |w| \leq 1. \end{cases}$$

By standard regularity results, we have

$$\|w_\varepsilon\|_{C^{1,\alpha}} \leq C,$$

where $C > 0$ is a universal constant. As a result, we assume that w_ε converges to some w_0 . Moreover w_0 satisfies the system

$$\begin{cases} \Delta w = \frac{(\beta-1)}{x_1} \left(-2 \frac{\partial w}{\partial x_1} \right) & \text{in } \{x_1 > 0\}. \\ \frac{\partial w}{\partial x_1} = 0 & \text{on } \partial\{u > 0\} \sim \{x_1 = 0\} \\ |w| \leq 1. \end{cases}$$

Then, there is a universal constant $C > 0$ such that

$$\|w_0\|_{C^{1,\alpha}(B_{\frac{1}{2}})} \leq C.$$

Now we choose

$$w = p \cdot x + O(r^{1+\alpha})$$

and have

$$\begin{aligned} u &= (x_1 + \varepsilon w_\varepsilon)_+ = (x_1 + \varepsilon w_0 + \varepsilon o(1))_+ \\ &= (x_1 + \varepsilon p \cdot x + \varepsilon o(1))_+ = \left(x \cdot e \pm \frac{1}{2} \varepsilon r_0 \right)_+ \end{aligned}$$

if we choose $0 < \varepsilon \ll 1$ and $0 < r_0 \ll 1$, then the desired properties hold. $\square$

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